

Confidence-Aware Multi-Task Graph Learning for Political Tweet Classification

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Abstract—Classifying political tweets requires separating the quality of a target label from the capacity of the model trained to predict it. We extend a leakage-audited graph learning project with a reproducible label cascade, frozen multilingual representations, larger residual networks, and measured runtime scaling. Of 7,053 tweets from 14 Macedonian political accounts, 6,537 have sufficient text for pseudo-labeling; two prompt formulations and confidence checks retain 1,169 for supervision. Acceptance measures teacher consistency, not independently verified truth. Ten neural configurations are evaluated over three initialization seeds on a common split that keeps duplicate normalized text together. A four-layer GraphSAGE model with 192 hidden units obtains 0.800 binary macro-F1, compared with 0.812 for a two-layer 96-unit model and 0.742 for text logistic regression. At full graph size, its warm optimizer step costs 4.08 times as much as the small model. Feature-only, edge-free, and single-task controls distinguish capacity from relational information. The pipeline makes uncertainty and computational cost observable while preserving the distinction between generated labels and factual misinformation.

Index Terms—graph neural networks, weak supervision, confidence filtering, social networks, runtime scaling

I. Introduction

Political communication contains factual claims, opinions, campaign slogans, accusations, and links to material outside a tweet. A classifier may assign all of these texts a label, but a numerical prediction alone does not establish whether a statement is true. This matters especially when another model generates the labels: a downstream network can learn the teacher’s preferences without learning factual verification.

The first version of this project compared GCN, GraphSAGE, and GAT for author identification and binary misinformation-label prediction. An audit revealed direct tweet–author edges, which encoded the author target. Removing those edges and author-profile-count features reduced the near-perfect identification scores. After correction, the strongest graph model was approximately tied with text logistic regression on generated misinformation labels. However, label confidence was not retained, pre-processing differed between graph and text models, and capacity and runtime were not studied systematically.

This revision addresses those weaknesses. It asks four questions: how can uncertain valid/invalid assignments be exposed; do multilingual representations and confidence filtering change performance on a common test set; does increasing graph-model width or depth help;

and what computational cost accompanies those changes? The contribution is an evaluated pipeline and controlled comparisons, rather than a new message-passing algorithm or a claim of automated fact-checking.

II. Background

GCN aggregates features with degree-normalized graph convolution [1]. GraphSAGE combines a node representation with an aggregated neighborhood representation [2]. GAT learns attention coefficients for neighboring nodes [3]. These encoders provide different ways to use retweeter neighborhoods, but all depend on an appropriate graph. A relation that reveals the target can invalidate an otherwise conventional evaluation.

Text features provide an essential comparison. TF-IDF encodes lexical frequencies, while multilingual sentence representations can capture related expressions with different wording. We use a frozen 384-dimensional multilingual MiniLM sentence model with attention-mask-aware mean pooling [4]. Feature extraction is a one-time preparation step, leaving the trainable graph models relatively inexpensive.

A multilingual natural language inference (NLI) model scores hypotheses associated with candidate classes [5]. Its outputs depend on class wording and available input evidence. Agreement and confidence are useful diagnostics of teacher behavior, but they do not replace human annotations or external evidence. Our binary task explicitly measures agreement with this teacher.

III. Dataset and Graph Audit

The archive contains 14 CSV files and 7,053 distinct tweet IDs. Tweets connect to account nodes through listed retweeters. Edges are made bidirectional and deduplicated; an account matching the tweet’s own author is excluded even if listed as a retweeter. The graph has 7,053 tweet nodes, 3,793 account nodes, and 114,314 directed edges. A post-construction assertion verifies zero direct authorship edges. Account nodes begin with zero features, and no trainable account-ID embedding is introduced.

The experiment is transductive: held-out tweet nodes, their text, and permitted relations are visible during message passing, but their labels do not enter the loss. This is classification in a known network, not prediction for new political accounts or future events. Retweeter activity

may occur after publication, so it is also not a real-time screening test.

Preparation removes URLs and account handles, normalizes whitespace, and preserves the original language. IDs are read as strings to avoid floating-point rounding. The earlier audit found 150 URL-only tweets. The revised evidence gate requires at least five word tokens after cleaning: 6,537 tweets qualify and 516 short or link-only texts are excluded from supervised training and evaluation. All nodes remain available as unlabeled graph context. Separately, 678 tweets have no retained retweeter edges and therefore no relational account context.

IV. Label-Generation Pipeline

A. Teacher and Prompt Views

The revised teacher is mDeBERTa-v3-base-mnli-xnli, loaded from a fixed revision in full precision [5]. We score original-language text without an additional translation stage. Its model card describes multilingual transfer from MNLI/XNLI training, but Macedonian-specific fact-checking accuracy is not established. The teacher and embedding model revisions are recorded in the manifest.

Each eligible tweet is evaluated against three hypotheses in each of two views. The first uses the original class words: “This example is valid/misleading/invalid.” The second uses descriptions: “This text contains a credible factual statement/a misleading statement/a false factual statement.” Entailment logits are normalized by softmax across candidates. The first view defines the fixed three-way target, whose most likely class is mapped to binary: valid is zero, and misleading or invalid is one. The second view checks agreement rather than redefining the test target.

Teacher input pairs and sentence-feature inputs are capped at 128 tokens. Long tweets may therefore lose trailing context, an additional limitation for claim interpretation. This length cap is fixed across candidates and recorded in the cache configuration.

B. Confidence and Abstention

Let $p_i^{(v)}(c)$ denote the normalized candidate score for class c in view v , and $b_i^{(v)}$ the merged hard three-way decision. The confidence in its binary vote is

$$q_i^{(v)} = \begin{cases} p_i^{(v)}(\text{valid}), & b_i^{(v)} = 0, \\ p_i^{(v)}(\text{misleading}) + p_i^{(v)}(\text{invalid}), & b_i^{(v)} = 1. \end{cases} \quad (1)$$

Acceptance requires sufficient text, equal binary votes, $\min_v q_i^{(v)} \geq 0.75$, and a gap of at least 0.15 between the two largest three-way scores in both views. These rules are fixed before downstream training. Generation is resumable and caches full score vectors rather than only the argmax.

The gate retains 17.9% of eligible examples; binary prompt agreement across eligible tweets is 74.1%. Table I reports stage counts. Merging two classes can increase positive-class confidence, so the post-filter class distribution is also recorded. Rejected labels are marked uncertain

TABLE I
Label cascade. Short/link-only texts have no supervised target; accepted rows pass both prompt and confidence checks.

Stage	Tweets	Valid	Misleading/invalid
All tweets	7,053	–	–
Eligible text	6,537	5,485	1,052
Accepted labels	1,169	722	447

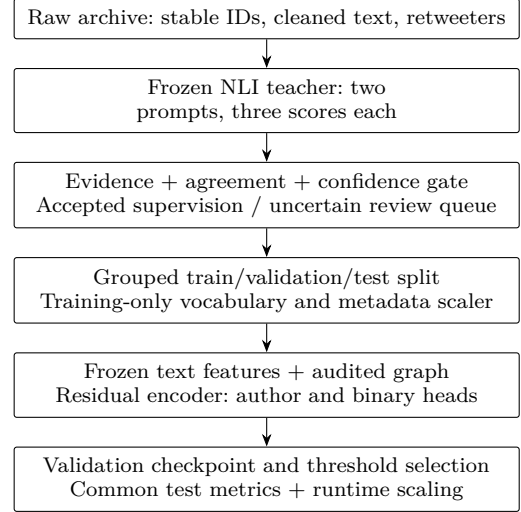


Fig. 1. Revised pipeline. The first teacher view fixes the target; acceptance changes binary training supervision rather than the test population.

and withheld from filtered binary training. Their first-view target is retained for teacher-agreement evaluation, and a separate review queue supports later human annotation.

Holding agreement and the 0.15 margin rule fixed, reducing the confidence threshold to 0.60 retains 35.9% of eligible examples; raising it to 0.90 retains 6.0%. This sensitivity audit exposes the coverage cost of a stricter gate. It does not estimate factual precision, and no alternative gate is selected using test performance.

V. Models and Evaluation

A. Partitions and Features

Equal case-folded cleaned text stays in one partition using stratified group folds. Outer and inner seeds are 42 and 43; stratification uses author and binary teacher class. The partitions contain 4,173 training, 1,038 validation, and 1,326 eligible test tweets. A group-intersection check prevents exact normalized duplicates from crossing partitions. It does not exclude paraphrases, related stories, or campaign themes.

The 256-term TF-IDF vocabulary and log-activity scaler are fitted on the complete training partition only. Numeric features comprise retweet, reply, and favorite counts; author follower/following counts are absent. The base representation has 259 dimensions. Adding 384 normalized frozen MiniLM dimensions gives 643 input

features. The encoder is applied independently without adaptation to the collection, so neither preprocessing statistics nor encoder training use held-out labels.

B. Architecture and Capacity

Every neural model projects its input to a hidden representation. A residual block applies GCN, GraphSAGE, GAT, or a feature-only linear transformation, adds the block input, then uses layer normalization, ReLU, and 0.25 dropout. Separate linear heads predict author and binary class. Small graph models have two blocks and 96 hidden units; GAT uses two averaged attention heads per block.

GraphSAGE capacity is increased in two steps. SAGE-wide uses two 192-unit blocks. SAGE-deep uses four 192-unit blocks with residual paths. Small, wide, and deep parameter counts are 100,816, 275,344, and 423,952, respectively. The deep model is designated in advance as the principal larger-model comparison.

The sequence isolates practical changes. SAGE-TFIDF uses lexical/activity features, and SAGE-small adds sentence features. SAGE-filtered keeps the small architecture but applies accepted-label supervision and confidence weighting. Width and depth increase after that step. MLP-control has the deep model’s rich features, accepted supervision, four blocks, and hidden width, but no graph. SAGE-no-edges retains the deep GraphSAGE parameterization while removing every edge. SAGE-single-task disables the author loss. These controls separate feature capacity, neighborhood information, and the auxiliary task.

C. Training and Operating Point

Models use AdamW with learning rate 0.002, weight decay 10^{-4} , gradient clipping at 2.0, at most 70 epochs, and patience 12. The primary loss is class-balanced binary cross-entropy with weights computed from available training targets. Filtered configurations average it over 735 accepted training rows, weighting each by minimum prompt confidence. The author loss continues to use all eligible training tweets, since screen names are independently observed. Joint models add author cross-entropy with weight 0.3.

Post-update, dropout-free validation binary macro-F1 selects the checkpoint. Its threshold is selected on validation data from 0.10 to 0.90 in steps of 0.05; ties favor proximity to 0.5. Neither epochs, thresholds, nor architectures are selected on test labels. Default-threshold metrics are saved alongside selected-threshold results.

Initialization seeds 42, 43, and 44 vary neural weights on the fixed split. Neural F1 values are means with sample standard deviations. Text logistic regression uses class-balanced unigram/bigram TF-IDF and the same validation threshold procedure; filtered text and majority predictors are controls. Every headline binary score uses all eligible test tweets, including abstained rows. Restricting testing to accepted tweets would change difficulty and could inflate apparent improvement.

TABLE II
Author identification on the revised grouped split. Mean $\pm$ sample SD over three seeds.

Encoder/configuration	Macro-F1
SAGE-TFIDF	0.602 $\pm$ 0.047
GCN-small	0.611 $\pm$ 0.040
SAGE-small	0.524 $\pm$ 0.062
GAT-small	0.640 $\pm$ 0.100
SAGE-filtered	0.431 $\pm$ 0.081
SAGE-wide	0.549 $\pm$ 0.072
SAGE-deep	0.384 $\pm$ 0.089
MLP-control	0.311 $\pm$ 0.041
SAGE-no-edges	0.329 $\pm$ 0.137

TABLE III
Binary teacher-label prediction on every eligible test tweet. Precision/recall refer to misleading or invalid; neural F1 is mean $\pm$ sample SD.

Model	Macro-F1	Prec.	Recall
Text-LR	0.742	0.578	0.544
Text-LR-filtered	0.734	0.569	0.525
Majority	0.458	0.000	0.000
SAGE-TFIDF	0.743 $\pm$ 0.014	0.518	0.645
GCN-small	0.818 $\pm$ 0.007	0.623	0.802
SAGE-small	0.812 $\pm$ 0.004	0.617	0.783
GAT-small	0.815 $\pm$ 0.004	0.626	0.783
SAGE-filtered	0.800 $\pm$ 0.005	0.554	0.881
SAGE-wide	0.802 $\pm$ 0.009	0.555	0.886
SAGE-deep	0.800 $\pm$ 0.005	0.565	0.846
MLP-control	0.789 $\pm$ 0.010	0.529	0.894
SAGE-no-edges	0.786 $\pm$ 0.003	0.527	0.889
SAGE-single-task	0.795 $\pm$ 0.008	0.557	0.838

VI. Predictive Results

A. Representation and Confidence Filtering

Tables II and III separate observed-author prediction from teacher-label agreement. SAGE-TFIDF achieves 0.743 binary macro-F1; adding sentence features gives 0.812. Corresponding author macro-F1 values are 0.602 and 0.524. This is a feature comparison within the revised setup, whereas historic scores used a different target procedure and split.

Confidence filtering and weighting without added capacity yields 0.800 binary macro-F1. Filtered text logistic regression obtains 0.734, versus 0.742 for its all-label fit. These comparisons expose both the potential benefit of avoiding uncertain supervision and the cost in available examples. Acceptance does not create verified labels and may preferentially retain easier or rhetorically distinctive text. The raw outputs also retain accepted/abstained test results, class-specific precision/recall, average precision, ROC-AUC, Brier scores, and confusion matrices.

Here the conservative gate does not improve aggregate F1 for either small GraphSAGE or text logistic regression. It also changes the positive-class share from 16.1% among eligible tweets to 38.2% among accepted labels. Thus confidence filtering is an abstention policy with a substantial coverage and selection-bias cost, not an empirically estab-

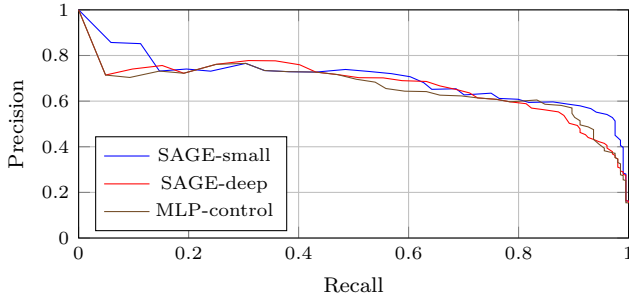


Fig. 2. Precision–recall curves for seed 42 on the common eligible test set. These describe agreement with teacher labels.

lished accuracy improvement. Rich text representations provide the clearer gain in this comparison.

B. Larger Models and Ablations

SAGE-wide and SAGE-deep reach 0.802 and 0.800 binary macro-F1. The deep model’s author macro-F1 is 0.384. Its binary F1 difference from SAGE-small is -0.0118 , with paired 95% bootstrap interval $[-0.0305, +0.0061]$; against the small filtered model it is -0.0004 with interval $[-0.0138, +0.0133]$. These distinguish the complete supervision/capacity combination from added capacity alone.

The MLP obtains 0.789, edge-free GraphSAGE 0.786, and the single-task graph model 0.795. The deep model’s difference from the MLP is $+0.0112$ with interval $[-0.0024, +0.0258]$. Improvement over a smaller GNN alone cannot establish that edges help: the non-graph controls are needed. Figure 2 shows the seed-42 precision–recall tradeoff beyond one threshold.

Intervals use 1,000 paired resamples of normalized-text groups in the test partition, recomputing the mean F1 difference across three seeds. They are conditional on this teacher and fixed split, not uncertainty from replacing the teacher, annotation procedure, dataset, or event. Three initialization seeds likewise check stability rather than deployment performance.

Increasing capacity does not establish a benefit in this setting: the deep model is approximately tied with filtered small GraphSAGE and trails the all-label small model despite its higher cost. The author scores are measured at checkpoints selected for binary validation F1, not at author-optimal checkpoints; early binary convergence can therefore leave the auxiliary author head undertrained. This protocol prioritizes the valid/invalid task requested in the revision and makes no claim of improved author identification.

All three paired intervals include zero. In particular, the positive deep-versus-MLP point estimate is not sufficient evidence of a reliable relational advantage. The appropriate conclusion is inconclusive evidence for added graph capacity under this teacher and split, rather than a claim that graph structure never helps.

TABLE IV
Full-graph CPU cost: parameter counts in thousands and median warm latency in milliseconds (seven repetitions).

Model	Params (k)	Train step	Inference
SAGE-small	100.8	133	44
GAT-small	101.6	333	117
SAGE-wide	275.3	290	90
SAGE-deep	424.0	541	169

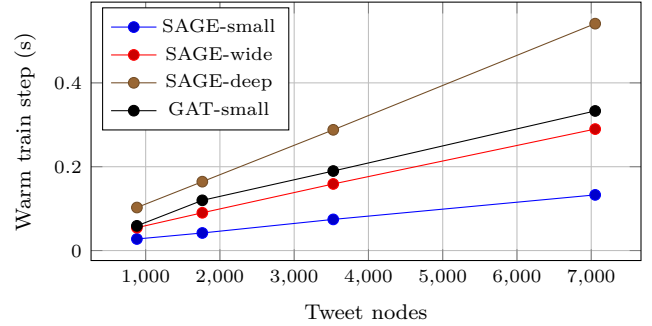


Fig. 3. Measured optimizer-step scaling on nested real tweet subgraphs plus their account neighbors. No synthetic tweets are added.

VII. Runtime and Scaling

Scaling benchmarks use the same eligible supervised rows for every configuration to isolate architectural cost, rather than the differing accepted-label counts used during predictive training. The full-size benchmark also preserves isolated account nodes.

Runtime is measured on a Windows laptop with an Intel Core i7-1255U CPU, four PyTorch threads, float32 GNN computation, and no GPU. Four nested subsets contain roughly one-eighth, one-quarter, one-half, and all real tweets plus their retained account neighbors. Each induced subgraph remaps account IDs. No synthetic or replicated tweets are added, and actual node/edge counts are saved.

The recorded environment uses PyTorch 2.8.0 and PyG 2.6.1 through the available PyTorch operators; optional compiled PyG acceleration extensions were unavailable. Backend and hardware differences can therefore change absolute timings on another machine.

For small, wide, and deep GraphSAGE and small GAT, we separately time the first optimizer step, discard two further warm-up steps, and record seven synchronized warm steps. A training step includes forward propagation, both supervised losses, backward propagation, and AdamW. It excludes validation, teacher inference, feature extraction, and I/O. Inference is measured over seven evaluation-mode forward passes. CPU work completes before reading the clock; CUDA is explicitly synchronized if rerun on GPU. Medians and interquartile ranges are saved rather than the fastest observation.

Table IV gives full-graph cost, and Fig. 3 shows the

trend with tweet count. Deep GraphSAGE costs 4.08 times small GraphSAGE per warm full-graph optimizer step, independently of early-stopped epoch counts. Total per-run training time including validation is also retained. Teacher throughput is measured separately across NLI pair batch sizes, including tokenization but excluding loading/downloads.

More tweets also change account and edge counts. The curve therefore measures observed subgraph scaling, not an asymptotic law at fixed degree. Sizes are bounded by this collection and cannot justify extrapolation to millions of tweets. GAT and GraphSAGE have different attention and aggregation costs, so parameter count alone does not predict runtime.

An isolated teacher benchmark uses 24 eligible tweets sampled with seed 42, three repetitions, and one prompt view (three NLI pairs per tweet). Pair batch sizes of 3, 12, and 24 achieve 2.09, 1.79, and 1.86 tweets per second, respectively. These warm measurements include tokenization and exclude model loading. The two-view label cascade requires twice the candidate evaluations, and variable sequence lengths affect throughput; these rates are not the latency of the complete downstream pipeline.

VIII. Discussion and Limitations

The revision improves auditability by recording teacher scores, screening insufficient text, exposing prompt disagreement, fitting training-only preprocessing, grouping duplicates, and measuring added capacity. These are process improvements even when a score comparison does not favor a larger model. They distinguish supervision changes from architecture changes and support interpretation beyond one accuracy number.

Factual validity remains the main unresolved limitation. The teacher does not retrieve linked articles, check dates, corroborate claims, or consult independent fact-checks. Criticism or subjective language may trigger its misleading class. The two prompts are correlated judgments from one model, and relative softmax confidence is not calibrated correctness. Outputs should be described as valid/invalid according to this procedure, with generated targets explicitly called pseudo-labels.

This revision changes the earlier XLM-R teacher to mDeBERTa, uses original-language inputs, and changes features and splitting. Historic binary F1 around 0.70 is not directly comparable to the new target. The within-revision ablations provide the relevant evidence. The original audit remains instructive, but its unaudited near-perfect author scores are not valid generalization evidence.

Other limits include one graph, one grouped random split, possible related-content overlap, and no temporal or

account-held-out test. Author accuracy cannot validate a weak binary label. Filtering can introduce selection bias and imbalance. A review queue with blank human-label/evidence fields supports independent annotation; automatic predictions are never counted as human verification. Human labels and temporal evaluation are the most valuable next steps toward a claim about real misinformation.

IX. Conclusion

The project now evaluates a complete labeling, modeling, and scaling pipeline. Ten neural configurations over three seeds include wider/deeper residual GraphSAGE and feature-only, edge-free, and single-task controls. Grouped partitions and a common eligible test set prevent duplicate leakage and changes in test difficulty. CPU scaling connects prediction scores to cost. The findings concern teacher-label agreement and author prediction; actual misinformation detection still requires independent annotation and evidence.

The measured outcome favors improved text features over added capacity: small GraphSAGE rises from 0.743 to 0.812 with multilingual representations, whereas deep filtered GraphSAGE achieves 0.800 at 4.08 times the warm training-step cost. Strict confidence filtering improves transparency about uncertainty but does not improve aggregate F1 here. This makes independent label validation a better next investment than simply enlarging the network.

Code and Reproducibility

The GitHub repository includes the revised Colab notebook, Python scripts, pinned model revisions, run histories, split indices, raw timing samples, and generated LaTeX tables/figures. Raw tweets, labels, weights, and review files are excluded. README commands regenerate labels, model comparisons, benchmarks, and numeric paper artifacts.

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